

Symmetrical Reality-Based Large-Scale Simulation of Embodied Agents for the Evolutionary Virtual World

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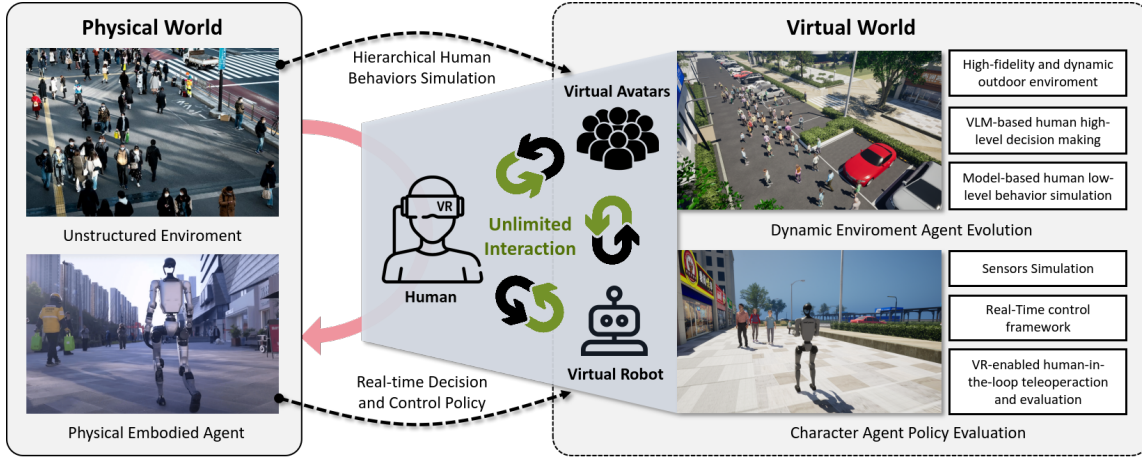


Figure 1: Overview of the Symmetric Reality framework connecting the physical world and a high-fidelity evolutionary virtual world, enabling mirrored human–robot interaction, scalable behavior evolution, and policy evaluation..

ABSTRACT

Autonomous robots face significant challenges in dynamic, unstructured, and socially complex outdoor environments. A key limitation is the lack of frameworks for evaluating perception, navigation, and cognition under realistic social dynamics. Leveraging the Symmetrical Reality (SR) paradigm, we present a large-scale SR-based simulation framework for embodied intelligence in dynamic social settings. It integrates a hierarchical human behavior model that combines Vision-Language Models for high-level decision-making with model-based low-level motion planning, enabling avatars to respond intelligently to robot behaviors. The framework further supports VR-enabled human-in-the-loop interaction, allowing users to act as virtual robots or humans for bidirectional interaction and rich data collection.

Index Terms: Symmetrical Reality, Human-Robot Social Interaction, Virtual Human.

1 INTRODUCTION AND RELATED WORKS

General-purpose robotics requires embodied agents capable of operating in human-centric, unstructured environments, yet real-world deployment remains challenging due to costly data collection, limited diversity, and safety risks [1].

Existing embodied-intelligence simulators inadequately capture environmental realism and human-centric dynamics. Early indoor platforms such as AI2-THOR [2] and VirtualHome [3] improve visual fidelity but are limited to static scenes, while newer human-centric simulators (e.g., Habitat 3.0 [4], Virtual Community [6])

support large-scale multi-agent activities yet still lack realistic human motion, social norms, and reciprocal interactions. Consequently, learned policies generalize poorly to unstructured, densely populated outdoor environments, motivating a unified framework for cognitively grounded, socially interactive simulation.

To address these limitations (Figure 1), Symmetrical Reality [5] tightly couples virtual and physical worlds, enabling large-scale embodied-agent simulation in continually evolving environments, where agents are trained and refined before safe real-world transfer.

2 PROBLEM FORMULATION

We study human–robot social interaction in a dynamic environment with N_h humans and one robot, abstracted as an evolutionary virtual world $\mathcal{E}$ in which agent states $s_{h_i}(t)$ and $s_r(t)$, as well as environmental factors (e.g., lighting, weather, and vegetation), evolve over time.

Environment Agents: Modeled as large-scale virtual human avatars, are assumed to follow a Markov Decision Process (MDP) governed by a Social Force Model (SFM) augmented with a Visual-Language Module (VLM):

$$s_{h_i}(t+1) \sim T_h(s_{h_i}(t), s_{-h_i}(t); \Theta), \quad (1)$$

where avatar actions are not represented as explicit action variables but are implicitly embedded in the human transition dynamics T_h via the SFM, which captures mutual interactions among humans and the robot. Here, $s_{-h_i}(t) = \{s_{h_j}(t)\}_{j \neq i} \cup \{s_r(t)\}$ denotes the states of all other human agents and the robot, and $\Theta = f_{\text{VLM}}(\text{image, map, prompt})$ represents interaction parameters conditioned on VLM outputs. We assume the simulator has full observability of the collective avatar states $s_h(t)$ and the character agent state $s_r(t)$.

Character Agent: Modeled as a socially aware virtual robot, it operates under a Partially Observable Markov Decision Process

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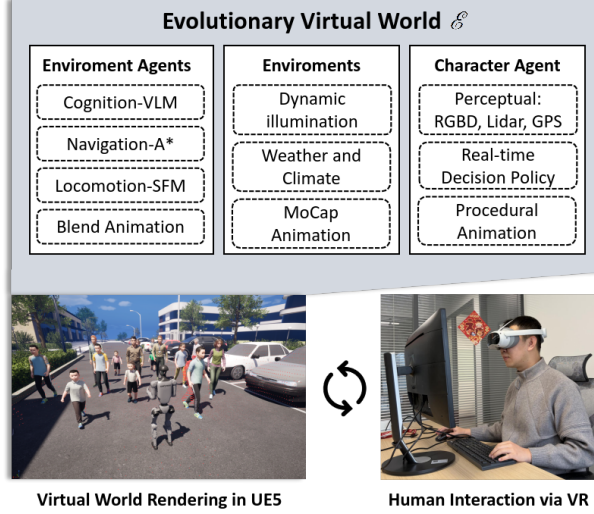


Figure 2: Evolutionary Virtual World framework with environment agents, environment modules, and character agents, plus virtual crowd simulation and user interaction.

(POMDP) due to sensor noise, occlusions, and the inherent unobservability of human intent. The agent maintains a belief $b_r(t)$ over the joint state $s(t) = \{s_r(t), s_h(t)\}$, which is recursively updated from its previous action $a_r(t-1)$ and *noisy, partial observations* $o_r(t)$ that provide only a limited, local view of the surrounding environment.

$$o_r(t) \sim \Omega(\cdot | s(t)) \quad (2)$$

Its policy $\pi_r : b_r(t) \mapsto a_r(t)$ aims to maximize expected task performance and social compliance:

$$\max_{\pi_r} \mathbb{E}_{\pi_r, T, \Omega} \left[\sum_{t=0}^T R(s_r(t), s_h(t), a_r(t)) \right]. \quad (3)$$

3 SYSTEM ARCHITECTURE AND IMPLEMENTATION

Our simulation framework is built upon Unreal Engine 5 (UE5), as illustrated in Figure 2, which provides high-fidelity rendering and a robust animation system for constructing the virtual world $\mathcal{E}$.

3.1 Hierarchical Environment Agent Simulation

We adopt a three-layer control hierarchy for virtual avatars in $\mathcal{E}$. A programmable crowd generator initializes N_h agents with diverse conditions, while adjusting VLM prompts or SFM parameters produces targeted crowd behaviors, yielding diverse scenarios for evaluating character agent policies.

High-Level Decision Making via VLM: A VLM modulates the SFM using semantic cues rather than rigid geometry. It infers a navigational target g_{global} from the global map and local observations, and interprets higher-level features (e.g., emotion, crowd density) to produce behavioral parameters θ_{SFM} . Thus, $\Theta = \{g_{\text{global}}, \theta_{\text{SFM}}\}$,

Table 1: Social Navigation Performance in Virtual World $\mathcal{E}$

Method	EFF	SR	COL	SN	Score
Human Teleop	0.89	1.00	0.95	0.88	0.93
Random	0.13	0.03	0.05	0.10	0.08
Smac A* + DWA	0.51	0.10	0.13	0.17	0.21
Smac A* + MPPI	0.73	0.60	0.25	0.31	0.43

allowing the agent to pursue its goal while adapting its navigation style without direct intervention in low-level control.

Mid-Level Navigation with A* Sampling: To avoid local minima, g_{global} is converted into a sequence of global path waypoints $P = \{w_0, \dots, w_n\}$ via A* on UE5’s NavMesh. Local waypoints from P are selected sequentially as the current SFM attractor w_{local} , guiding the agent toward the goal.

Low-Level Locomotion and Animation with SFM: The SFM integrates parameters to compute a target position, tracked by a PD controller at the engine level, decoupling computation from rendering. The resulting velocity $\vec{v}_i$ drives the kinematic controller, while UE5’s BlendSpace interpolates MoCap walk/run cycles to align physical motion with skeletal animation.

3.2 Character Agent Implementation:

The character agent simulates real-world robot deployment by perceiving the environment through a 360° LiDAR and RGB-D camera to form observations $o_r(t)$ for policy π_r , executing continuous velocity commands (v, ω) , and using full body IK-based procedural animation to maintain accurate ground contact on terrain while producing a natural gait.

3.3 VR-based Human-in-the-Loop Interaction

Compared with the physical world, VR enables humans to be inserted into any stage of the perception–decision–action loop within $\mathcal{E}$. Through immersive telepresence and real-time bidirectional control, humans can take on the roles of observers, supervisors, demonstrators, or direct teleoperators, interacting with both virtual avatars and virtual robot under fully controllable, repeatable, and safe conditions. This flexibility enables tasks like behavior evaluation, policy debugging, demonstration collection, and user-experience studies, while preserving ecological validity without physical constraints. Agents validated in this virtual environment can then be safely deployed in the physical world.

4 EXPERIMENT AND DISCUSSION

As a representative case, we consider a social navigation task in our environment $\mathcal{E}$. As Table 1 shows, classical planning methods that treat humans as moving obstacles (e.g., Smac A* + DWA, Smac A* + MPPI) underperform compared to human teleoperation. The evaluation uses four metrics—**EFF** (efficiency, 0.2), **SR** (success rate, 0.2), **COL** (collision count, 0.3), and **SN** (social norm compliance, 0.3)—combined into a weighted overall score. While classical methods handle geometric avoidance and basic trajectory optimization, they fail to model human intent, social spacing, yielding, or group interactions, resulting in inefficient and socially non-compliant navigation in dense environments, highlighting the need for richer social cognition in robot decision-making.

REFERENCES

- [1] R. Firoozi, J. Tucker, S. Tian, et al. Foundation models in robotics: Applications, challenges, and the future. *The International Journal of Robotics Research*, 44:701–739, 2023. 1
- [2] E. Kolve, R. Mottaghi, W. Han, et al. Ai2-thor: An interactive 3d environment for visual ai. *ArXiv*, abs/1712.05474, 2017. 1
- [3] X. Puig, K. K. Ra, M. Boben, et al. Virtualhome: Simulating household activities via programs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8494–8502, 2018. 1
- [4] X. Puig, E. Undersander, A. Szot, et al. Habitat 3.0: A co-habitat for humans, avatars and robots. *ArXiv*, abs/2310.13724, 2023. 1
- [5] Z. Zhang, Z. Zhang, Z. Jiao, et al. On the emergence of symmetrical reality. In *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pages 639–649. IEEE, 2024. 1
- [6] Q. Zhou, H. Zhang, X. Lin, et al. Virtual community: An open world for humans, robots, and society. *ArXiv*, abs/2508.14893, 2025. 1